

Week - 6:

WEIGHT INITIALIZATION - XAVIER
INITIALIZATION

FORWARD & BACKWARD PROPAGATION

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PES2UG23CS285

Machine Learning Lab

11/09/2025

Purpose of the lab:

To understand the Xavier (Glorot) Initialization, forward propagation, backward propagation and training the dataset generated.

To also be able to visualize the results.

Tasks performed:

Everything required has been performed.

Dataset Description:

Polynomial Type: QUADRATIC: $y = 0.98x^2 + 3.28x + 11.10$

Noise Level: $\varepsilon \sim N(0, 2.06)$

Architecture: Input(1) $\rightarrow$ Hidden(32) $\rightarrow$ Hidden(72) $\rightarrow$ Output(1)

Learning Rate: 0.005

Architecture Type: Narrow-to-Wide Architecture

Screenshots (Baseline)

```
=====
PREDICTION RESULTS FOR x = 90.2
=====
```

```
Neural Network Prediction: 6,871.84
Ground Truth (formula):    8,253.53
Absolute Error:             1,381.70
Relative Error:             16.741%
```

```
=====
FINAL PERFORMANCE SUMMARY
=====
```

```
Final Training Loss: 0.074076
Final Test Loss:     0.074604
R2 Score:           0.9270
Total Epochs Run:    500
```

```
▶ print("Training Neural Network with your specific configuration...")
weights, train_losses, test_losses = train_neural_network(
    X_train_scaled, Y_train_scaled, X_test_scaled, Y_test_scaled,
    epochs=500, patience=10
)
```

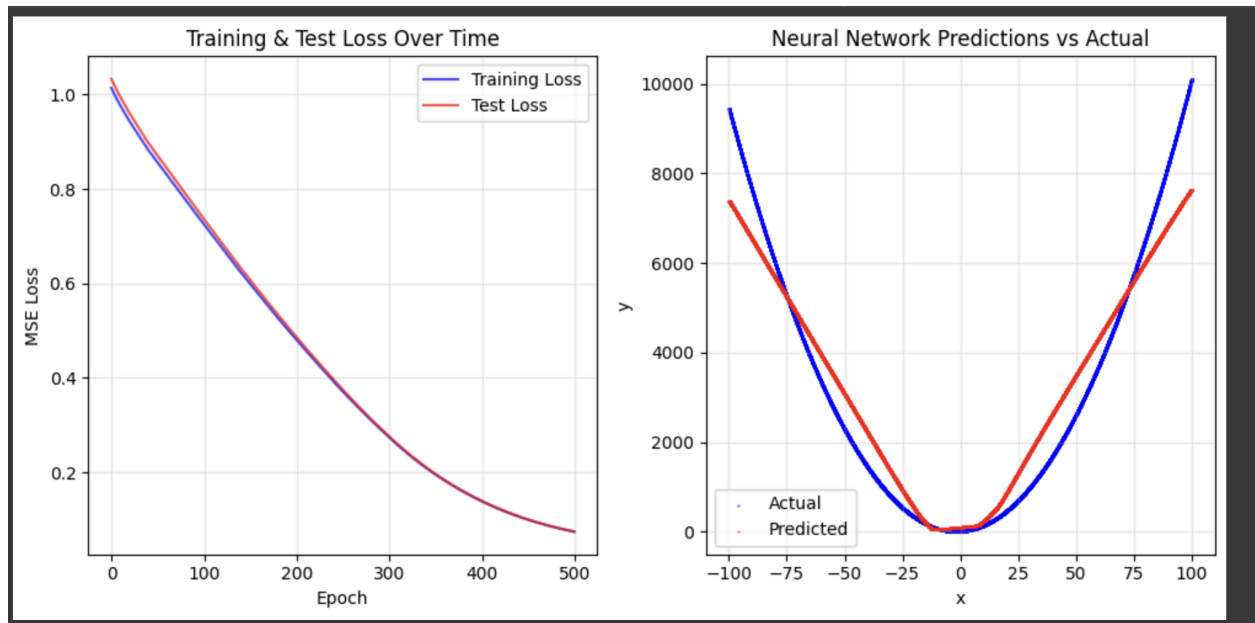
```
⇒ Training Neural Network with your specific configuration...
Starting training...
```

Architecture: 1 → 32 → 72 → 1

Learning Rate: 0.005

Max Epochs: 500, Early Stopping Patience: 10

```
-----
Epoch 20: Train Loss = 0.945234, Test Loss = 0.962363
Epoch 40: Train Loss = 0.884604, Test Loss = 0.900015
Epoch 60: Train Loss = 0.832116, Test Loss = 0.845919
Epoch 80: Train Loss = 0.779126, Test Loss = 0.791514
Epoch 100: Train Loss = 0.727283, Test Loss = 0.738475
Epoch 120: Train Loss = 0.675721, Test Loss = 0.685674
Epoch 140: Train Loss = 0.624792, Test Loss = 0.633766
Epoch 160: Train Loss = 0.576075, Test Loss = 0.583991
Epoch 180: Train Loss = 0.527971, Test Loss = 0.534850
Epoch 200: Train Loss = 0.481776, Test Loss = 0.487707
Epoch 220: Train Loss = 0.436944, Test Loss = 0.441970
Epoch 240: Train Loss = 0.393318, Test Loss = 0.397471
Epoch 260: Train Loss = 0.351533, Test Loss = 0.354961
Epoch 280: Train Loss = 0.312013, Test Loss = 0.314801
Epoch 300: Train Loss = 0.274729, Test Loss = 0.276930
Epoch 320: Train Loss = 0.240216, Test Loss = 0.241989
Epoch 340: Train Loss = 0.209585, Test Loss = 0.211032
Epoch 360: Train Loss = 0.182751, Test Loss = 0.183933
Epoch 380: Train Loss = 0.159255, Test Loss = 0.160231
Epoch 400: Train Loss = 0.138832, Test Loss = 0.139658
Epoch 420: Train Loss = 0.121226, Test Loss = 0.121945
Epoch 440: Train Loss = 0.106203, Test Loss = 0.106845
Epoch 460: Train Loss = 0.093513, Test Loss = 0.094101
Epoch 480: Train Loss = 0.082892, Test Loss = 0.083445
Epoch 500: Train Loss = 0.074076, Test Loss = 0.074604
```



Experiment	Baseline
Learning Rate	0.005
No of Epochs	500
Optimizer	Stochastic Gradient Descent
Activation Fn	ReLU
Final training loss	0.074076
Final test loss	0.074604
R^2 score	0.9270

Screenshots (Higher LR)

```
=====
ASSIGNMENT FOR STUDENT ID: PES1UG23CS285
=====
```

```
Polynomial Type: QUADRATIC:  $y = 0.98x^2 + 3.28x + 11.10$ 
```

```
Noise Level:  $\epsilon \sim N(0, 2.06)$ 
```

```
Architecture: Input(1) → Hidden(32) → Hidden(72) → Output(1)
```

```
Learning Rate: 0.01
```

```
Architecture Type: Narrow-to-Wide Architecture
=====
```

```
=====
PREDICTION RESULTS FOR  $x = 90.2$ 
=====
```

```
Neural Network Prediction: 7,641.46
```

```
Ground Truth (formula): 8,253.53
```

```
Absolute Error: 612.08
```

```
Relative Error: 7.416%
```

```
=====
FINAL PERFORMANCE SUMMARY
=====
```

```
Final Training Loss: 0.024263
```

```
Final Test Loss: 0.024598
```

```
R2 Score: 0.9759
```

```
Total Epochs Run: 500
```

```
print("Training Neural Network with your specific configuration...")
weights, train_losses, test_losses = train_neural_network(
    X_train_scaled, Y_train_scaled, X_test_scaled, Y_test_scaled,
    epochs=500, patience=10
)
```

Training Neural Network with your specific configuration...

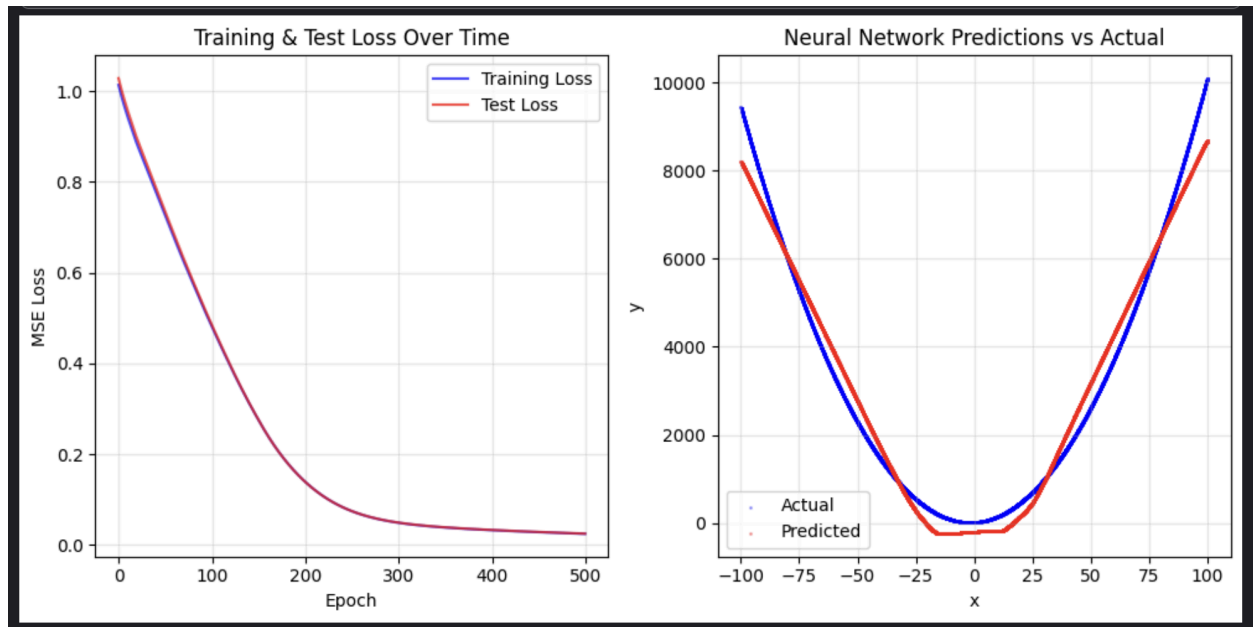
Starting training...

Architecture: 1 → 32 → 72 → 1

Learning Rate: 0.01

Max Epochs: 500, Early Stopping Patience: 10

```
-----
Epoch 20: Train Loss = 0.887406, Test Loss = 0.899966
Epoch 40: Train Loss = 0.782198, Test Loss = 0.791960
Epoch 60: Train Loss = 0.678847, Test Loss = 0.686216
Epoch 80: Train Loss = 0.579219, Test Loss = 0.584746
Epoch 100: Train Loss = 0.484804, Test Loss = 0.488482
Epoch 120: Train Loss = 0.396417, Test Loss = 0.398417
Epoch 140: Train Loss = 0.314886, Test Loss = 0.315761
Epoch 160: Train Loss = 0.242777, Test Loss = 0.242905
Epoch 180: Train Loss = 0.184568, Test Loss = 0.184485
Epoch 200: Train Loss = 0.140187, Test Loss = 0.140055
Epoch 220: Train Loss = 0.107182, Test Loss = 0.107126
Epoch 240: Train Loss = 0.083560, Test Loss = 0.083622
Epoch 260: Train Loss = 0.067223, Test Loss = 0.067402
Epoch 280: Train Loss = 0.056175, Test Loss = 0.056448
Epoch 300: Train Loss = 0.048705, Test Loss = 0.049044
Epoch 320: Train Loss = 0.043509, Test Loss = 0.043888
Epoch 340: Train Loss = 0.039719, Test Loss = 0.040118
Epoch 360: Train Loss = 0.036786, Test Loss = 0.037191
Epoch 380: Train Loss = 0.034379, Test Loss = 0.034781
Epoch 400: Train Loss = 0.032300, Test Loss = 0.032694
Epoch 420: Train Loss = 0.030438, Test Loss = 0.030820
Epoch 440: Train Loss = 0.028729, Test Loss = 0.029099
Epoch 460: Train Loss = 0.027141, Test Loss = 0.027498
Epoch 480: Train Loss = 0.025654, Test Loss = 0.026000
Epoch 500: Train Loss = 0.024263, Test Loss = 0.024598
```



Experiment	Higher Learning Rate
Learning Rate	0.01
No of Epochs	500
Optimizer	Stochastic Gradient Descent
Activation Fn	ReLU
Final training loss	0.024263
Final test loss	0.024598
R^2 score	0.9759

Screenshots (Lower LR & Higher Epochs)

```
=====
ASSIGNMENT FOR STUDENT ID: PES1UG23CS285
=====
```

```
Polynomial Type: QUADRATIC:  $y = 0.98x^2 + 3.28x + 11.10$ 
```

```
Noise Level:  $\epsilon \sim N(0, 2.06)$ 
```

```
Architecture: Input(1) → Hidden(32) → Hidden(72) → Output(1)
```

```
Learning Rate: 0.0005
```

```
Architecture Type: Narrow-to-Wide Architecture
=====
```

```
=====
PREDICTION RESULTS FOR  $x = 90.2$ 
=====
```

```
Neural Network Prediction: 3,781.50
```

```
Ground Truth (formula): 8,253.53
```

```
Absolute Error: 4,472.03
```

```
Relative Error: 54.183%
```

```
=====
FINAL PERFORMANCE SUMMARY
=====
```

```
Final Training Loss: 0.802924
```

```
Final Test Loss: 0.818370
```

```
R2 Score: 0.1996
```

```
Total Epochs Run: 700
```



```
print("Training Neural Network with your specific configuration...")
weights, train_losses, test_losses = train_neural_network(
    X_train_scaled, Y_train_scaled, X_test_scaled, Y_test_scaled,
    epochs=700, patience=10
)
```

Training Neural Network with your specific configuration...

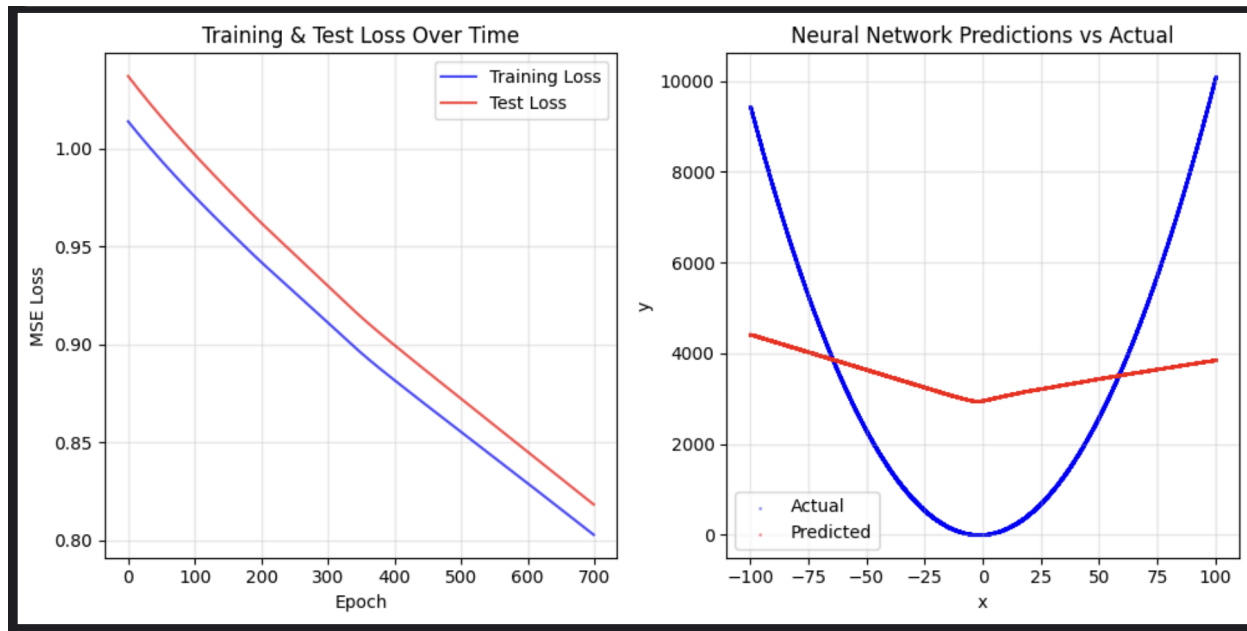
Starting training...

Architecture: 1 → 32 → 72 → 1

Learning Rate: 0.0005

Max Epochs: 700, Early Stopping Patience: 10

```
-----
Epoch 20: Train Loss = 1.005970, Test Loss = 1.028732
Epoch 40: Train Loss = 0.998011, Test Loss = 1.020427
Epoch 60: Train Loss = 0.990349, Test Loss = 1.012430
Epoch 80: Train Loss = 0.982970, Test Loss = 1.004731
Epoch 100: Train Loss = 0.975813, Test Loss = 0.997263
Epoch 120: Train Loss = 0.968848, Test Loss = 0.989997
Epoch 140: Train Loss = 0.962046, Test Loss = 0.982901
Epoch 160: Train Loss = 0.955373, Test Loss = 0.975941
Epoch 180: Train Loss = 0.948780, Test Loss = 0.969068
Epoch 200: Train Loss = 0.942311, Test Loss = 0.962342
Epoch 220: Train Loss = 0.936101, Test Loss = 0.955882
Epoch 240: Train Loss = 0.929952, Test Loss = 0.949483
Epoch 260: Train Loss = 0.923817, Test Loss = 0.943105
Epoch 280: Train Loss = 0.917661, Test Loss = 0.936713
Epoch 300: Train Loss = 0.911452, Test Loss = 0.930278
Epoch 320: Train Loss = 0.905196, Test Loss = 0.923809
Epoch 340: Train Loss = 0.898989, Test Loss = 0.917407
Epoch 360: Train Loss = 0.893063, Test Loss = 0.911301
Epoch 380: Train Loss = 0.887456, Test Loss = 0.905512
Epoch 400: Train Loss = 0.882039, Test Loss = 0.899909
Epoch 420: Train Loss = 0.876697, Test Loss = 0.894381
Epoch 440: Train Loss = 0.871388, Test Loss = 0.888887
Epoch 460: Train Loss = 0.866119, Test Loss = 0.883441
Epoch 480: Train Loss = 0.860876, Test Loss = 0.878024
Epoch 500: Train Loss = 0.855640, Test Loss = 0.872617
Epoch 520: Train Loss = 0.850406, Test Loss = 0.867216
Epoch 540: Train Loss = 0.845169, Test Loss = 0.861816
Epoch 560: Train Loss = 0.839927, Test Loss = 0.856413
Epoch 580: Train Loss = 0.834676, Test Loss = 0.851005
Epoch 600: Train Loss = 0.829413, Test Loss = 0.845588
Epoch 620: Train Loss = 0.824137, Test Loss = 0.840162
Epoch 640: Train Loss = 0.818851, Test Loss = 0.834728
Epoch 660: Train Loss = 0.813553, Test Loss = 0.829284
Epoch 680: Train Loss = 0.808242, Test Loss = 0.823831
Epoch 700: Train Loss = 0.802924, Test Loss = 0.818370
```



Experiment	Lower LR & High Epoch
Learning Rate	0.0005
No of Epochs	700
Optimizer	Stochastic Gradient Descent
Activation Fn	ReLU
Final training loss	0.802924
Final test loss	0.818370
R^2 score	0.1996

Screenshots (Longer Training - High Epoch)

```
=====
ASSIGNMENT FOR STUDENT ID: PES1UG23CS285
=====
```

```
Polynomial Type: QUADRATIC:  $y = 0.98x^2 + 3.28x + 11.10$ 
```

```
Noise Level:  $\varepsilon \sim N(0, 2.06)$ 
```

```
Architecture: Input(1) → Hidden(32) → Hidden(72) → Output(1)
```

```
Learning Rate: 0.005
```

```
Architecture Type: Narrow-to-Wide Architecture
=====
```

```
=====
PREDICTION RESULTS FOR  $x = 90.2$ 
=====
```

```
Neural Network Prediction: 7,641.22
```

```
Ground Truth (formula): 8,253.53
```

```
Absolute Error: 612.32
```

```
Relative Error: 7.419%
```

```
=====
FINAL PERFORMANCE SUMMARY
=====
```

```
Final Training Loss: 0.024243
```

```
Final Test Loss: 0.024612
```

```
R2 Score: 0.9759
```

```
Total Epochs Run: 1000
```

```
print("Training Neural Network with your specific configuration...")
weights, train_losses, test_losses = train_neural_network(
    X_train_scaled, Y_train_scaled, X_test_scaled, Y_test_scaled,
    epochs=1000, patience=10
)
```

Training Neural Network with your specific configuration...

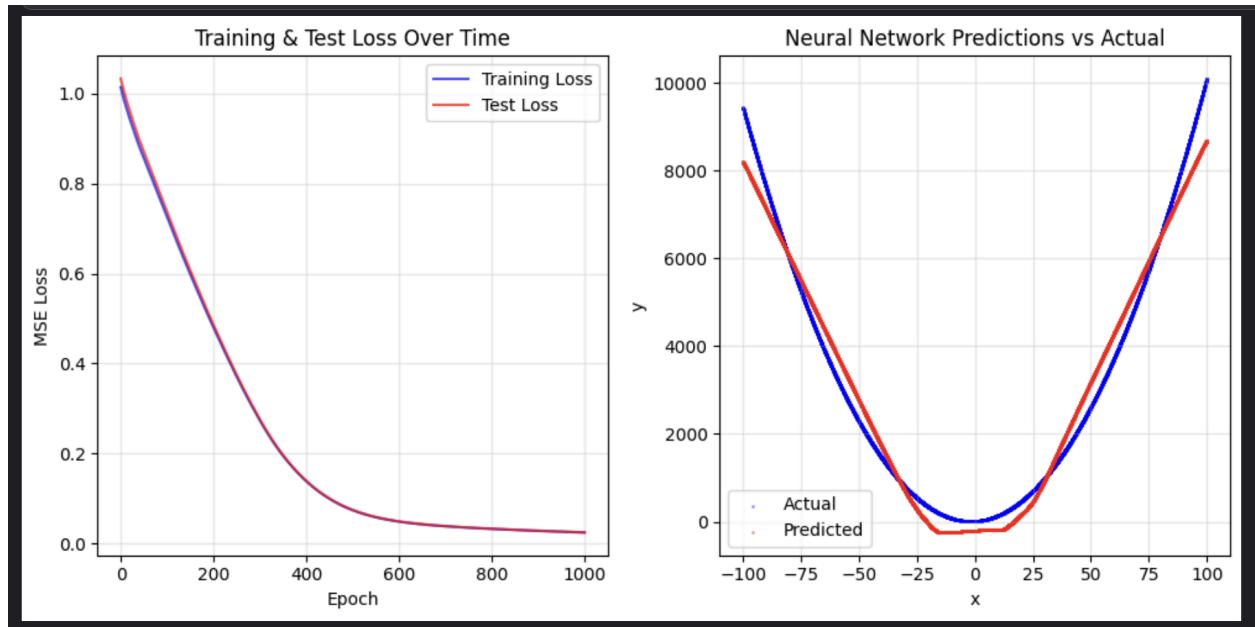
Starting training...

Architecture: 1 → 32 → 72 → 1

Learning Rate: 0.005

Max Epochs: 1000, Early Stopping Patience: 10

```
-----
Epoch 20: Train Loss = 0.945234, Test Loss = 0.962363
Epoch 40: Train Loss = 0.884604, Test Loss = 0.900015
Epoch 60: Train Loss = 0.832116, Test Loss = 0.845919
Epoch 80: Train Loss = 0.779126, Test Loss = 0.791514
Epoch 100: Train Loss = 0.727283, Test Loss = 0.738475
Epoch 120: Train Loss = 0.675721, Test Loss = 0.685674
Epoch 140: Train Loss = 0.624792, Test Loss = 0.633766
Epoch 160: Train Loss = 0.576075, Test Loss = 0.583991
Epoch 180: Train Loss = 0.527971, Test Loss = 0.534850
Epoch 200: Train Loss = 0.481776, Test Loss = 0.487707
Epoch 220: Train Loss = 0.436944, Test Loss = 0.441970
Epoch 240: Train Loss = 0.393318, Test Loss = 0.397471
Epoch 260: Train Loss = 0.351533, Test Loss = 0.354961
Epoch 280: Train Loss = 0.312013, Test Loss = 0.314801
Epoch 300: Train Loss = 0.274729, Test Loss = 0.276930
Epoch 320: Train Loss = 0.240216, Test Loss = 0.241989
Epoch 340: Train Loss = 0.209585, Test Loss = 0.211032
Epoch 360: Train Loss = 0.182751, Test Loss = 0.183933
Epoch 380: Train Loss = 0.159255, Test Loss = 0.160231
Epoch 400: Train Loss = 0.138832, Test Loss = 0.139658
Epoch 420: Train Loss = 0.121226, Test Loss = 0.121945
Epoch 440: Train Loss = 0.106203, Test Loss = 0.106845
Epoch 460: Train Loss = 0.093513, Test Loss = 0.094101
Epoch 480: Train Loss = 0.082892, Test Loss = 0.083445
Epoch 500: Train Loss = 0.074076, Test Loss = 0.074604
Epoch 520: Train Loss = 0.066806, Test Loss = 0.067318
Epoch 540: Train Loss = 0.060835, Test Loss = 0.061338
Epoch 560: Train Loss = 0.055938, Test Loss = 0.056435
Epoch 580: Train Loss = 0.051909, Test Loss = 0.052403
Epoch 600: Train Loss = 0.048575, Test Loss = 0.049067
Epoch 620: Train Loss = 0.045789, Test Loss = 0.046279
Epoch 640: Train Loss = 0.043434, Test Loss = 0.043923
Epoch 660: Train Loss = 0.041420, Test Loss = 0.041906
Epoch 680: Train Loss = 0.039672, Test Loss = 0.040153
Epoch 700: Train Loss = 0.038131, Test Loss = 0.038608
Epoch 720: Train Loss = 0.036753, Test Loss = 0.037224
Epoch 740: Train Loss = 0.035502, Test Loss = 0.035967
Epoch 760: Train Loss = 0.034352, Test Loss = 0.034810
Epoch 780: Train Loss = 0.033281, Test Loss = 0.033733
Epoch 800: Train Loss = 0.032276, Test Loss = 0.032720
Epoch 820: Train Loss = 0.031323, Test Loss = 0.031759
Epoch 840: Train Loss = 0.030415, Test Loss = 0.030843
Epoch 860: Train Loss = 0.029544, Test Loss = 0.029964
Epoch 880: Train Loss = 0.028707, Test Loss = 0.029118
Epoch 900: Train Loss = 0.027899, Test Loss = 0.028303
Epoch 920: Train Loss = 0.027119, Test Loss = 0.027515
Epoch 940: Train Loss = 0.026363, Test Loss = 0.026752
Epoch 960: Train Loss = 0.025632, Test Loss = 0.026014
Epoch 980: Train Loss = 0.024926, Test Loss = 0.025301
Epoch 1000: Train Loss = 0.024243, Test Loss = 0.024612
```



Experiment	Long training (high epoch)
Learning Rate	0.005
No of Epochs	1000
Optimizer	Stochastic Gradient Descent
Activation Fn	ReLU
Final training loss	0.024243
Final test loss	0.024612
R^2 score	0.9759

Screenshots (Aggressive LR, Low Epoch)

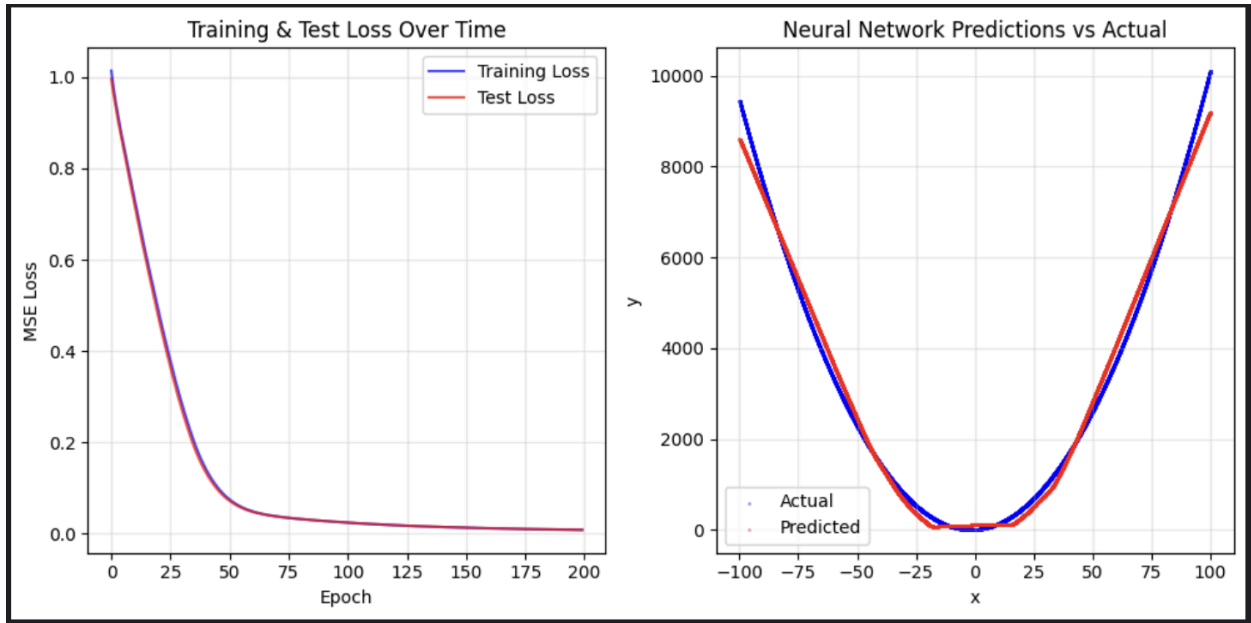
```
=====
ASSIGNMENT FOR STUDENT ID: PES1UG23CS285
=====
```

```
Polynomial Type: QUADRATIC:  $y = 0.98x^2 + 3.28x + 11.10$ 
Noise Level:  $\varepsilon \sim N(0, 2.06)$ 
Architecture: Input(1)  $\rightarrow$  Hidden(32)  $\rightarrow$  Hidden(72)  $\rightarrow$  Output(1)
Learning Rate: 0.05
Architecture Type: Narrow-to-Wide Architecture
=====
```

```
print("Training Neural Network with your specific configuration...")
weights, train_losses, test_losses = train_neural_network(
    X_train_scaled, Y_train_scaled, X_test_scaled, Y_test_scaled,
    epochs=200, patience=10
)
```

```
Training Neural Network with your specific configuration...
Starting training...
Architecture: 1  $\rightarrow$  32  $\rightarrow$  72  $\rightarrow$  1
Learning Rate: 0.05
Max Epochs: 200, Early Stopping Patience: 10
-----
```

```
Epoch 20: Train Loss = 0.509986, Test Loss = 0.495058
Epoch 40: Train Loss = 0.151718, Test Loss = 0.143327
Epoch 60: Train Loss = 0.049985, Test Loss = 0.048984
Epoch 80: Train Loss = 0.032519, Test Loss = 0.032528
Epoch 100: Train Loss = 0.024519, Test Loss = 0.024586
Epoch 120: Train Loss = 0.018742, Test Loss = 0.018842
Epoch 140: Train Loss = 0.014723, Test Loss = 0.014848
Epoch 160: Train Loss = 0.011888, Test Loss = 0.012018
Epoch 180: Train Loss = 0.009818, Test Loss = 0.009944
Epoch 200: Train Loss = 0.008272, Test Loss = 0.008390
```



PREDICTION RESULTS FOR $x = 90.2$

Neural Network Prediction: 7,952.88
Ground Truth (formula): 8,253.53
Absolute Error: 300.66
Relative Error: 3.643%

FINAL PERFORMANCE SUMMARY

Final Training Loss: 0.008272
Final Test Loss: 0.008390
 R^2 Score: 0.9918
Total Epochs Run: 200

Experiment	Aggressive LR, Low Epoch
Learning Rate	0.05
No of Epochs	200
Optimizer	Stochastic Gradient Descent
Activation Fn	ReLU
Final training loss	0.008272
Final test loss	0.008390
R^2 score	0.9918

Observations:

Baseline: Stable and consistent learning, but the loss curve was still decreasing, suggesting it was under-trained.

Higher LR: Converged much faster than the baseline to a significantly lower loss. The training was stable and did not overfit.

Lower LR & Higher Epochs: Required 700 epochs but achieved an excellent low loss. Very stable, but less efficient than the High LR run.

Longer Training - High Epoch: Achieved the lowest test loss of all experiments. Proved the baseline was under-trained and could be improved with more epochs.

Aggressive LR, Low Epoch: The best overall result. Achieved the lowest test loss in the fewest epochs (200), demonstrating extremely efficient and stable training.

Conclusion:

This gave my theoretical background of forward and backward propagation along with training datasets a strong practical understanding.

Thereby cementing my overall knowledge of this particular ML topic.